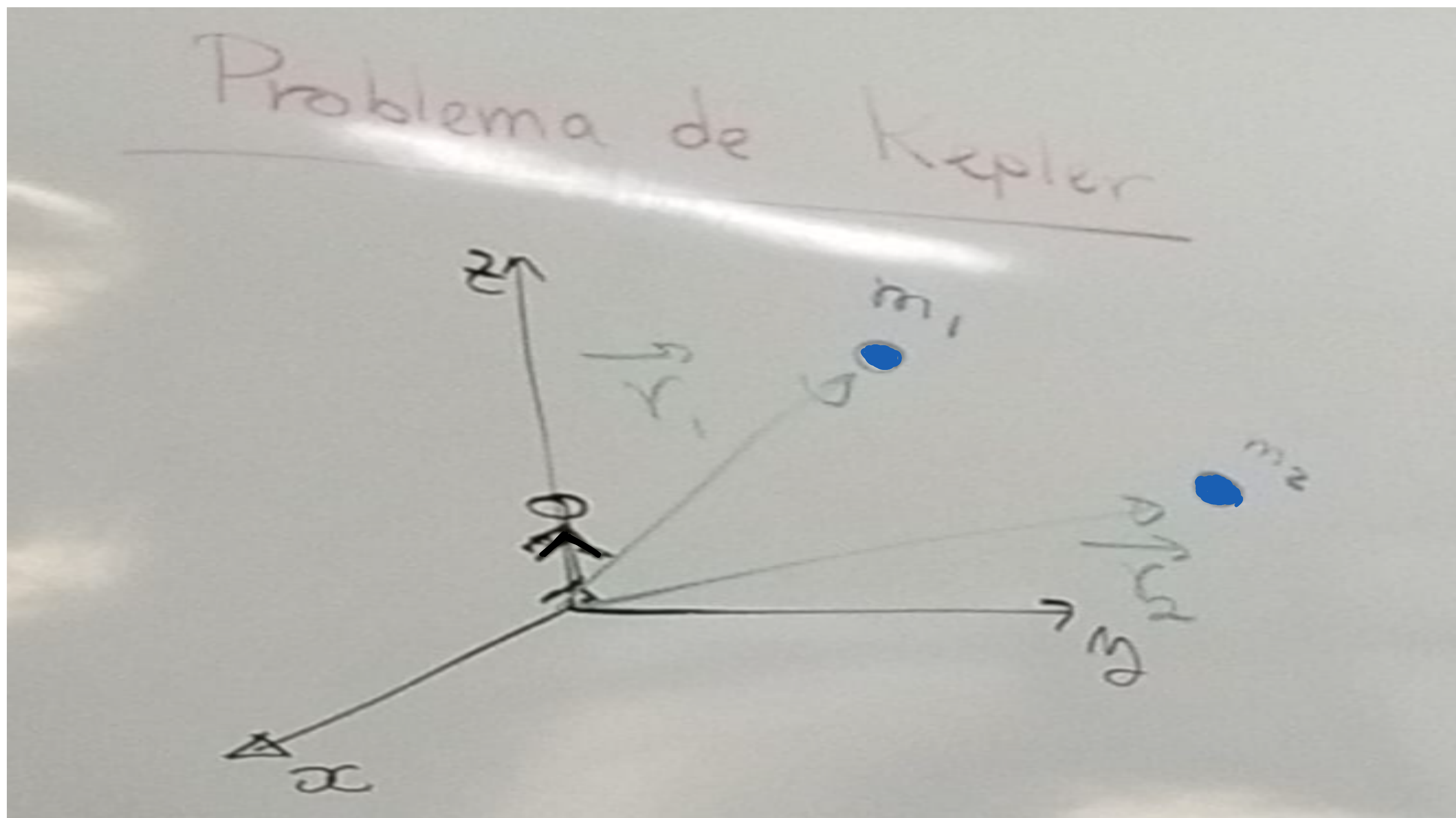


Problema de Kepler (No podemos reprobar este parcial)



$$\vec{F} = m \vec{a} = m \frac{d^2 \vec{r}}{dt^2}$$

- Calcular el valor de la fuerza es considerablemente arduo.

Como punto de partida utilizamos las siguientes coordenadas (vector de posición).

$$\vec{r}_1 = x_1 \hat{x} + y_1 \hat{y} + z_1 \hat{z} \quad \left. \vphantom{\vec{r}_1} \right\} \text{Partícula 1 tiene 3 grados de libertad}$$

$$\vec{r}_2 = x_2 \hat{x} + y_2 \hat{y} + z_2 \hat{z} \quad \left. \vphantom{\vec{r}_2} \right\} \text{También 3 grados de libertad}$$

$$\rightarrow \text{Ecuaciones de movimiento} \quad \left. \vphantom{\text{Ecuaciones de movimiento}} \right\} n = 3N - K = 3(2) - 0 = 6$$

$$\rightarrow \{x_1, y_1, z_1, x_2, y_2, z_2\}$$

Usando el cálculo de la Lagrangiana, tenemos:

$$L = T - U$$

Tenemos...

$$T = \frac{1}{2} m_1 \dot{\vec{r}}_1^2 + \frac{1}{2} m_2 \dot{\vec{r}}_2^2$$

$$L = \frac{1}{2} m_1 \dot{\vec{r}}_1^2 + \frac{1}{2} m_2 \dot{\vec{r}}_2^2$$

$$L = \frac{1}{2} m_1 (\dot{x}_1^2 + \dot{y}_1^2 + \dot{z}_1^2) + \frac{1}{2} m_2 (\dot{x}_2^2 + \dot{y}_2^2 + \dot{z}_2^2)$$

Euler-Lagrange

$$\left\{ \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0 \right\}_{i=1}^{n=6}$$

Tenemos

$$q_1 = x_1, \quad q_2 = y_1, \quad q_3 = z_1, \quad q_4 = x_2, \quad q_5 = y_2, \quad q_6 = z_2$$

$$\textcircled{i=1} \quad \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_1} \right) - \frac{\partial L}{\partial q_1} = 0$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_1} \right) - \frac{\partial L}{\partial x_1} = 0$$

$$\frac{d}{dt}(m_1 \dot{x}_1) = m_1 \ddot{x}_1 = 0$$

$$\left. \begin{array}{l} \textcircled{i=2} \quad m_1 \ddot{y}_1 = 0 \\ \textcircled{i=3} \quad m_1 \ddot{z}_1 = 0 \end{array} \right\}$$

Tenemos...

$$m_2 \ddot{x}_2 = 0 ; m_2 \ddot{y}_2 = 0 ; m_2 \ddot{z}_2 = 0$$

$$\ddot{\vec{r}}_1 = \vec{0} \Rightarrow \vec{r}_1(t) = \vec{r}_{01} + \vec{v}_{01} t$$

$$\ddot{\vec{r}}_2 = \vec{0} \Rightarrow \vec{r}_2(t) = \vec{r}_{02} + \vec{v}_{02} t$$

A photograph of a handwritten page showing the derivation of particle motion. At the top, the equations $m_2 \ddot{x}_2 = 0 ; m_2 \ddot{y}_2 = 0 ; m_2 \ddot{z}_2 = 0$ are written. Below them, a box contains the acceleration equations $\ddot{\vec{r}}_1 = \vec{0}$ and $\ddot{\vec{r}}_2 = \vec{0}$. Arrows point from these to the final position equations $\vec{r}_1(t) = \vec{r}_{01} + \vec{v}_{01} t$ and $\vec{r}_2(t) = \vec{r}_{02} + \vec{v}_{02} t$, which are also boxed. The equations are written in a clear, legible hand.

(x_1, y_1, z_1) no están en la Lagrangiana, entonces eso nos

dice que:

$$p_{x_1} = cte ; p_{y_1} = cte ; p_{z_1} = cte ; p_{x_2} = cte ; p_{y_2} = cte$$

$$p_{z_2} = cte.$$

Así tenemos

$$\vec{p}_1 = \vec{c} \vec{t} e ; \quad \vec{p}_2 = \vec{c} \vec{t} e$$

$$\vec{L}_1 = \vec{r}_1 \times \vec{p}_1 ; \quad \vec{L}_2 = \vec{r}_2 \times \vec{p}_2$$

$$\vec{L} = \vec{r}_1 \times \vec{p}_1 + \vec{r}_2 \times \vec{p}_2$$

$$\frac{\partial \mathcal{L}}{\partial t} = 0$$

Definimos lo siguiente:

$$\{ q_1 = x_1, \quad q_2 = y_1, \quad q_3 = z_1 ; \quad q_4 = x_2, \quad q_5 = y_2, \quad q_6 = z_2 \}$$

$$x \equiv x_2 - x_1 ; \quad y \equiv y_2 - y_1 ; \quad z \equiv z_2 - z_1$$

$$X \equiv \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2} ; \quad Y \equiv \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2}$$

↑
mayúscula

↑
mayúscula

$$Z = \frac{m_1 z_1 + m_2 z_2}{m_1 + m_2}$$

↑
mayúscula

Jacobiano de la
transformación diferente de
0

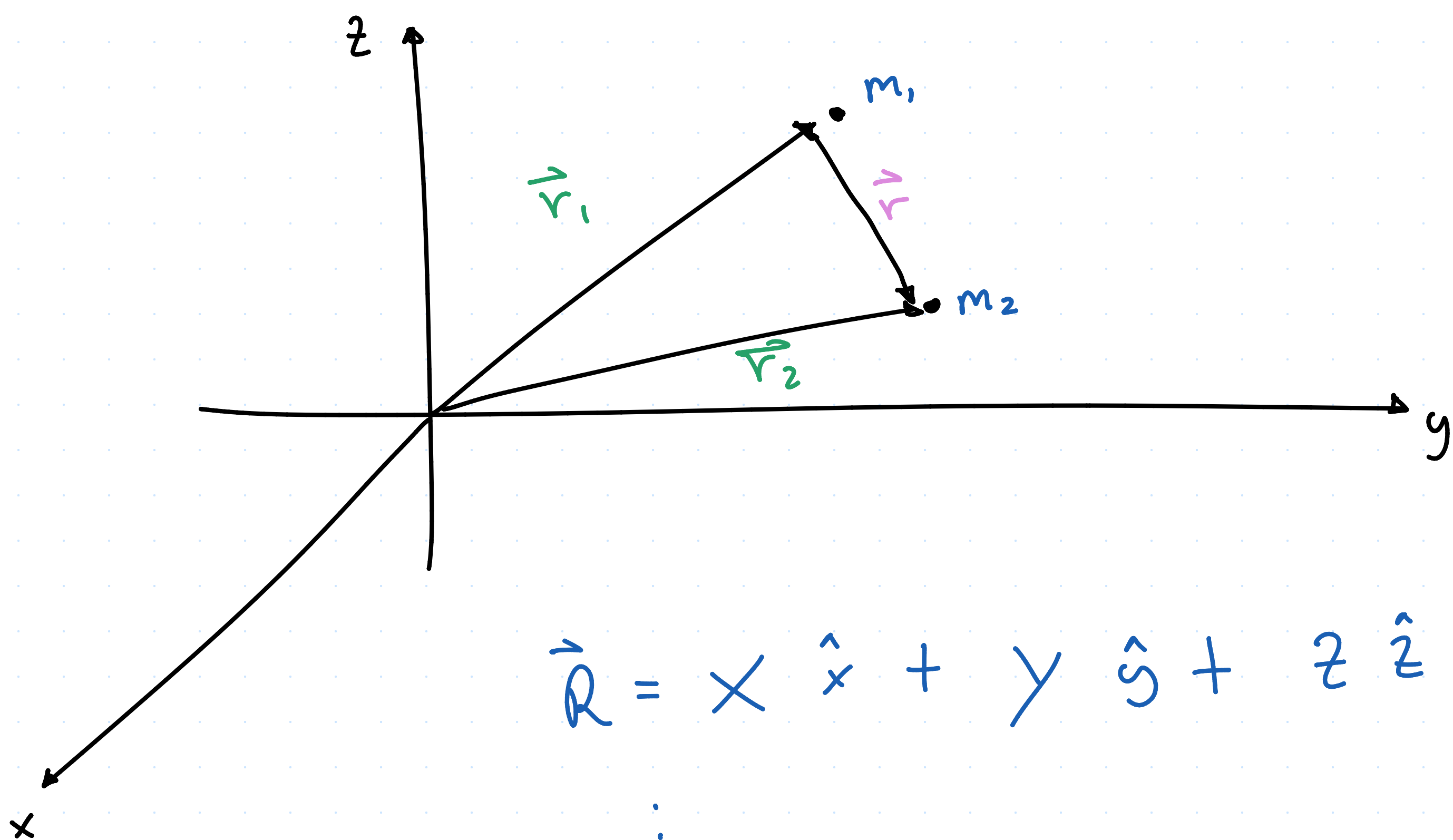
$J \neq 0$ (comprobar el
J de esta transform.)

Estas coordenadas (x, y, z) etiquetan a todo $\mathbb{R}^3$ de este

Problema.

Tenemos

$$\vec{r} = x \hat{x} + y \hat{y} + z \hat{z} = \vec{r}_2 - \vec{r}_1$$



$$\vec{R} = x \hat{x} + y \hat{y} + z \hat{z}$$

$$= \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2}$$

} centro de masas

$$M \equiv m_1 + m_2$$

Sustituimos

Tenemos

$$\vec{r} = \vec{r}_2 - \vec{r}_1$$

→

$$\vec{r}_2 = \vec{r} + \vec{r}_1$$

Tenemos

Sustituimos

$$\vec{Q} = \frac{m_1}{M} \vec{r}_1 + \frac{m_2}{M} (\vec{r} + \vec{r}_1)$$

$$\vec{Q} = \left(\frac{m_1}{M} + \frac{m_2}{M} \right) \vec{r}_1 + \frac{m_2}{M} \vec{r}$$

$$\vec{Q} = \left(\frac{m_1 + m_2}{M} \right) \vec{r}_1 + \frac{m_2}{M} \vec{r}$$

Sustituimos

$$\vec{Q} = \vec{r}_1 + \frac{m_2}{M} \vec{r} \Rightarrow \vec{r}_1 = \vec{Q} - \frac{m_2}{M} \vec{r}$$

$$\vec{r}_2 = \vec{Q} + \frac{m_1}{M} \vec{r}$$

vector relativo
vector centro de masas

Tenemos

$$\vec{r}_2 = \vec{r} + \vec{Q} - \frac{m_2}{M} \vec{r}$$

$$= \left(1 - \frac{m_2}{M} \right) \vec{r} + \vec{Q}$$

$$= \left(\frac{M}{M} - \frac{m_2}{M} \right) \vec{r} + \vec{Q}$$

Finalmente

Recordar que

Estamos trabajando con un sistema con dos partículas libres.

$$\vec{r}_2 = \vec{Q} + \frac{m_1}{M} \vec{r}$$

Tenemos

$$\dot{\vec{r}}_1 = \dot{\vec{R}} - \frac{m_2}{M} \dot{\vec{r}}$$

$$\dot{\vec{r}}_1^2 = \left(\dot{\vec{R}} - \frac{m_2}{M} \dot{\vec{r}} \right) \cdot \left(\dot{\vec{R}} - \frac{m_2}{M} \dot{\vec{r}} \right)$$

$$\vdots$$

$$= \dot{\vec{R}}^2 - 2 \frac{m_2}{M} \dot{\vec{R}} \cdot \dot{\vec{r}} + \frac{m_2^2}{M^2} \dot{\vec{r}}^2$$

Ahora para la partícula 2:

$$\dot{\vec{r}}_2 = \dot{\vec{R}} + \frac{m_1}{M} \dot{\vec{r}}$$

↑
velocidad
del centro
de masas

$$\dot{\vec{r}}_2^2 = \dot{\vec{R}}^2 + 2 \frac{m_1}{M} \dot{\vec{R}} \cdot \dot{\vec{r}} + \frac{m_1^2}{M^2} \dot{\vec{r}}^2$$

Tenemos

$$L = \frac{1}{2} m_1 \left[\dot{\vec{R}}^2 - 2 \frac{m_2}{M} \dot{\vec{R}} \cdot \dot{\vec{r}} + \frac{m_2^2}{M^2} \dot{\vec{r}}^2 \right] + \frac{1}{2} m_2 \left[\dot{\vec{R}}^2 + 2 \frac{m_1}{M} \dot{\vec{R}} \cdot \dot{\vec{r}} + \frac{m_1^2}{M^2} \dot{\vec{r}}^2 \right]$$

*

Tenemos...

$$L = \frac{1}{2} m_1 \dot{\vec{r}}_1^2 + \frac{1}{2} m_2 \dot{\vec{r}}_2^2$$

$$\vec{r}_1 = \vec{R}_1 - \frac{m_2}{M} \vec{r} \quad ; \quad \vec{r}_2 = \vec{R} + \frac{m_1}{M} \vec{r} \quad ; \quad \dot{\vec{r}}_1 = \dot{\vec{0}} \quad ; \quad \dot{\vec{r}}_2 = \dot{\vec{0}}$$

Tenemos

*

$$= \frac{1}{2} (m_1 + m_2) \dot{R}^2 + \frac{1}{2} \left(\frac{m_1 m_2^2}{M^2} + \frac{m_2 m_1^2}{M^2} \right) \dot{r}^2$$

$$L = \frac{1}{2} M \dot{R}^2 + \frac{1}{2} \frac{m_1 m_2}{M} \left(\frac{m_2 + m_1}{M} \right) \dot{r}^2$$

$$L = \frac{1}{2} M \dot{R}^2 + \frac{1}{2} m \dot{r}^2$$

Recordando que :

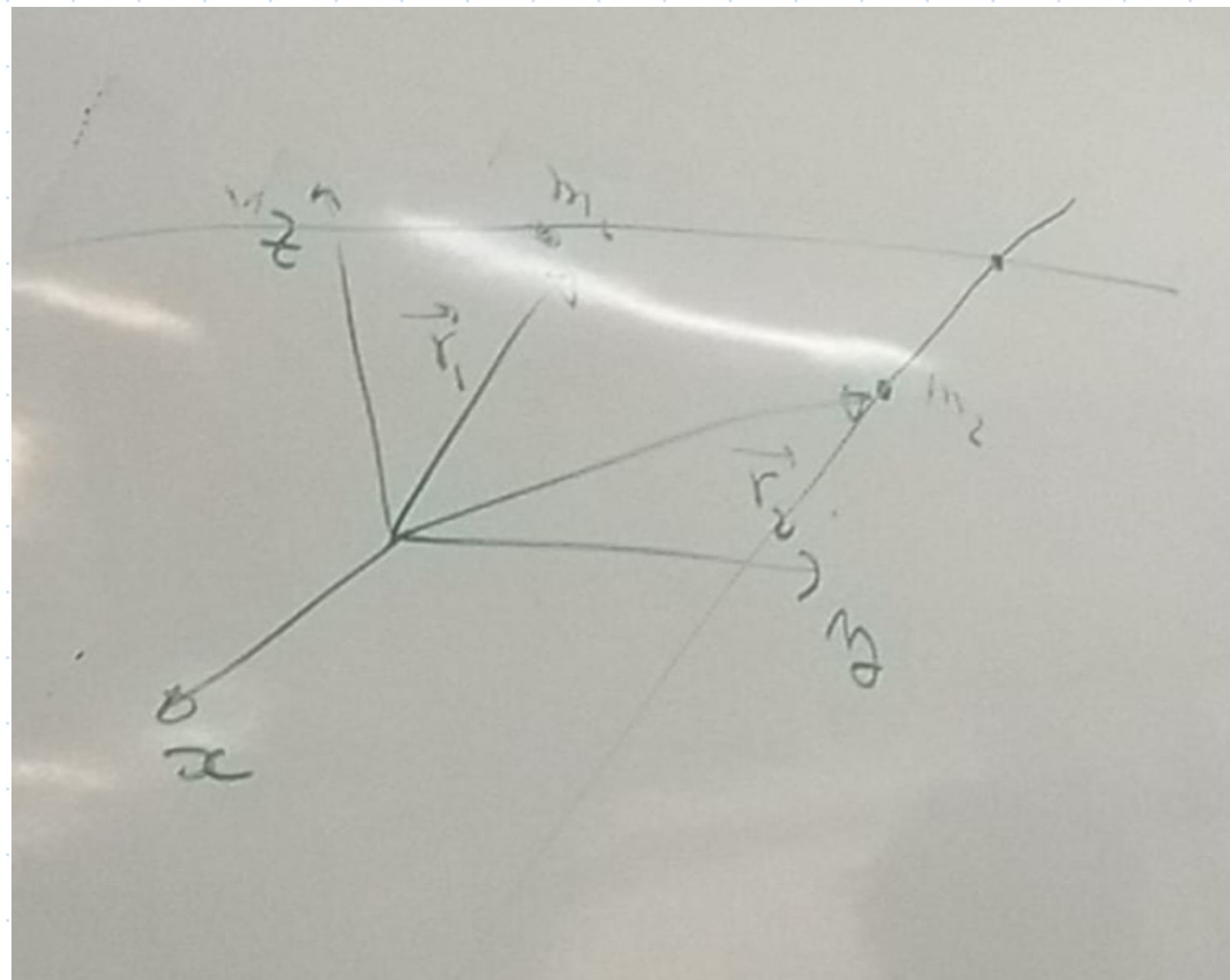
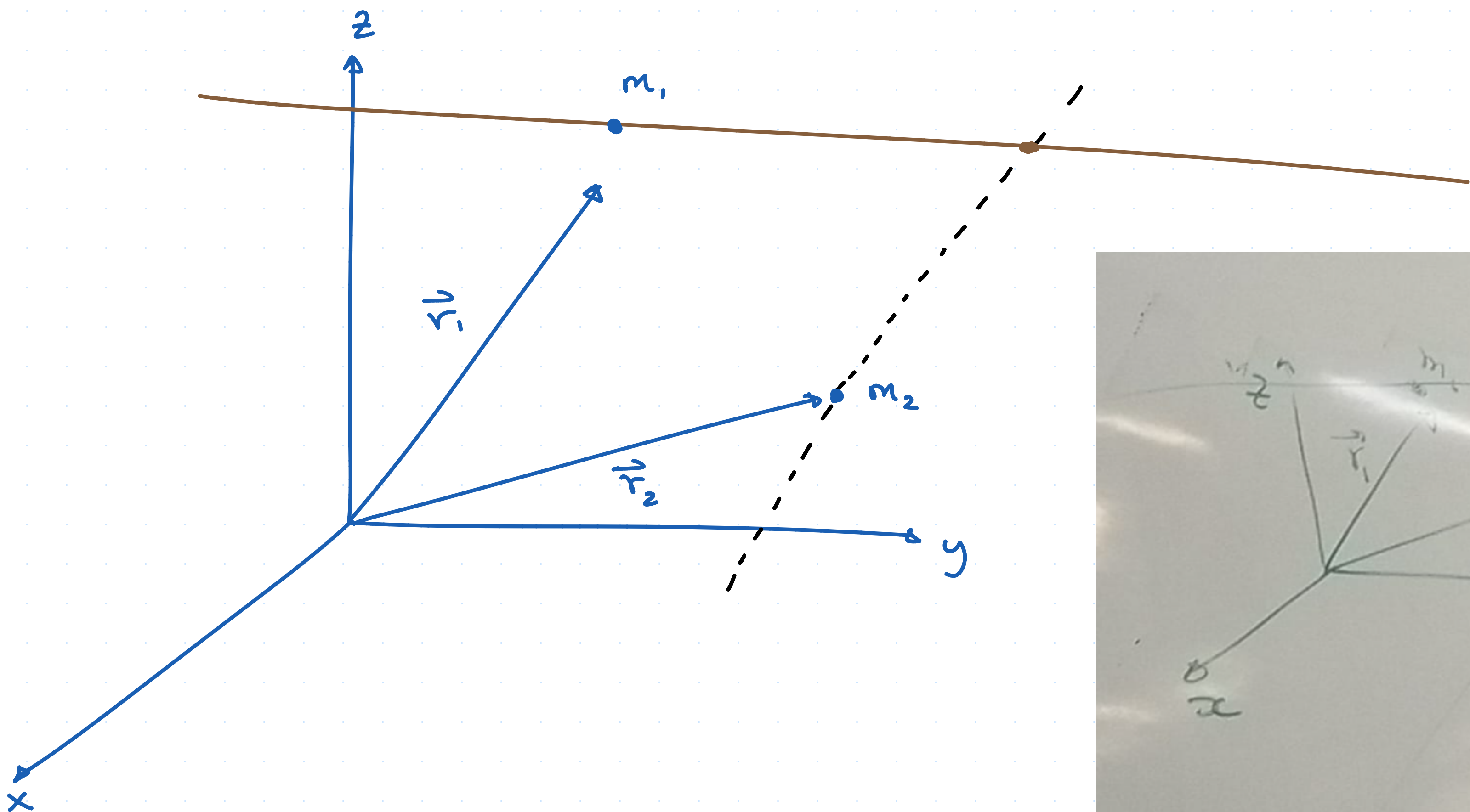
$$M = m_1 + m_2 ;$$

Masa reducida $\Rightarrow$

$$m \equiv \frac{m_1 m_2}{m_1 + m_2}$$

$$L = \frac{1}{2} m_1 \dot{r}_1^2 + \frac{1}{2} m_2 \dot{r}_2^2$$

Recordando que estamos trabajando en sistema de 2 partículas libres.



Transformación usada:

$$\vec{r}_1 = \vec{R} - \frac{m_2}{M} \vec{r} ; \quad \vec{r}_2 = \vec{R} + \frac{m_1}{M} \vec{r}$$

$$\underline{m} = \frac{m_0 m_T}{m_0 + m_T}$$

$$\frac{m_T}{m_0} \ll 1$$

$$= \frac{m_T}{1 + \frac{m_T}{m_0}} \approx m_T$$

$$1 + \frac{m_T}{m_0}$$

Serie geométrica

$$\frac{1}{1+x} = \sum_{n=0}^{\infty} (-x)^n = 1 ;$$

$$|x| < 1 ;$$

$$m \approx m_T$$

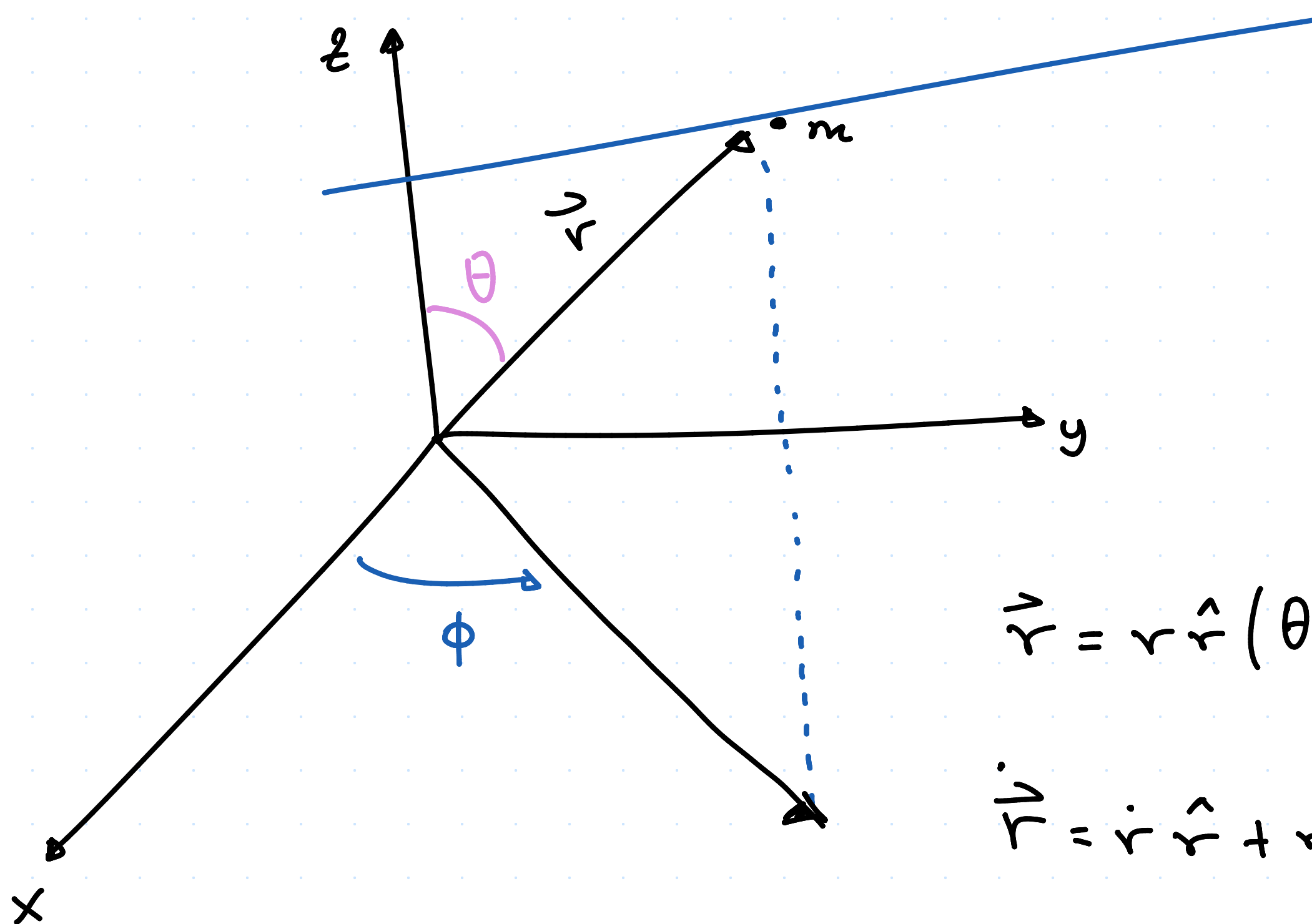
Ahora realizamos el análisis en coordenadas esféricas.

Tenemos

$$\mathcal{L} = \frac{1}{2} m \dot{\vec{r}}^2 ;$$

↑

Nos han dado esta lagrangiana.



$$\vec{r} = r \hat{r}(\theta, \phi)$$

$$\dot{\vec{r}} = \dot{r} \hat{r} + r \frac{d\hat{r}}{dt}$$

Necesitamos

determinar $\vec{r}(t)$

Tenemos

$$\dot{\vec{r}} = \dot{r} \hat{r} + r \frac{\partial \hat{r}}{\partial \theta} \dot{\theta} + r \frac{\partial \hat{r}}{\partial \phi} \dot{\phi}$$

$$\dot{\vec{r}} = \dot{r} \hat{r} + r \dot{\theta} \hat{\theta} + r \dot{\phi} \sin \theta \hat{\phi}$$

Tenemos

$$\mathcal{J} = \frac{1}{2} m \left[\dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \dot{\phi}^2 \sin^2 \theta \right]$$

Tenemos

$$q_1 = r ; \quad q_2 = \theta, \quad q_3 = \phi$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{J}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{J}}{\partial q_i} = 0 ; \quad \{i\}_{i=1}^{n=3}$$

Tenemos

$$q_1 = r ;$$

$$\frac{\partial \mathcal{J}}{\partial r} = m (r \dot{\theta}^2 + r \dot{\phi}^2 \sin^2 \theta) = m r (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2)$$

$$\frac{\partial \mathcal{J}}{\partial \dot{r}} = m \dot{r}$$

$$m \dot{r} - m r (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) = 0$$

$$\frac{\partial \mathcal{J}}{\partial \theta} = m r^2 \dot{\phi}^2 \sin \theta \cos \theta$$

$$\frac{\partial \mathcal{J}}{\partial \dot{\theta}} = m r^2 \dot{\theta}$$

$$\mathcal{J} = \frac{1}{2} m \left[\dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \dot{\phi}^2 \sin^2 \theta \right]$$

$$\frac{d}{dt}(mr^2\dot{\theta}) - mr^2\dot{\phi}^2 \sin\theta \cos\theta = 0$$

$$\frac{\partial \mathcal{L}}{\partial \phi} = 0; \quad \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = mr^2\dot{\phi} \sin^2\theta$$

$$\frac{d}{dt}(mr^2\dot{\theta} \sin^2\theta) = 0$$

$$\ddot{x} = 0$$

$$\ddot{y} = 0$$

$$\ddot{z} = 0$$

objetivo: hallar las ecuaciones de movimiento para (r, θ, ϕ)

$$\vec{L} = \vec{r} \times \vec{p}$$

$$\begin{aligned} \frac{d\vec{L}}{dt} &= \dot{\vec{r}} \times \vec{p} + \vec{r} \times \dot{\vec{p}} \\ &= \dot{\vec{r}} \times m\dot{\vec{r}} + \vec{r} \times \vec{F} = 0 = \vec{0} \end{aligned}$$

$\vec{L}$ es un vector constante

↑
momento angular
orbital

